

Aaron Tom Varghese

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SUMMARY

Recent Computer Science graduate with a strong foundation in software engineering principles and hands-on exposure to full-stack development. Brings practical experience building MVC application and developing hardware-software integrations through a recent internship and comprehensive university projects. Eager to apply strong problem-solving skills and a passion for continuous learning in a dynamic tech environment.

EDUCATION

- **TKM Institute of Technology** Kollam, Kerala
Bachelor of Technology in Computer Science & Engineering Oct 2022- May 2026
 - Cumulative GPA: 8.16/10.0
 - Relevant Coursework: Data Structures and Algorithms, Formal Language and Automata Theory, Database Management System, Python (Programming Language), Object Oriented Programming in Java, Operating Systems

WORK EXPERIENCE

- **Software Developer Intern** Kochi, Kerala
Voyon Technology Services Jan 2026 – Apr 2026
 - Engineered a mock **MVC application** for an Attendance System using **C#** and **.NET**, utilizing **ADO.NET** for efficient data access and management.
 - Architected and integrated an **AI Agent** feature to automate monitoring of employee work milestones and birthday anniversaries, improving internal engagement tracking.
 - Implemented comprehensive **Test Automation** using **Cypress**, conducting end-to-end testing on the MVC platform to ensure robust functionality and system stability.

CERTIFICATIONS

- **Python for Data Science**
NPTEL IIT Madras
- **Programming in Python**
Meta (Coursera)
- **Artificial Intelligence and Machine Learning**
SMEC Institute, Kaloor, Kochi

UNIVERSITY PROJECT

- **Rash Driving Detection System for Public Buses** Aug 2025
 - Architected a **hybrid edge-cloud framework** using **Raspberry Pi 5** and **Python** to detect aggressive driving (harsh braking, tailgating) in real-time
 - Developed a **1D Kalman Filter** for sensor fusion, combining **MPU-6050 IMU** data with **GPS streams** to maintain accurate velocity estimates even during signal dropouts.
 - Integrated **Computer Vision** via **OpenCV** (Haar Cascades/Contours) for tailgating detection and **ultrasonic sensors** for side-proximity (close overtaking) monitoring.
 - Engineered a **resilient data pipeline** using a **Store-and-Forward** mechanism with **SQLite**, ensuring zero data loss during network disruptions through automatic background synchronization.
 - Built a full-stack monitoring ecosystem including a **Flask backend**, **Socket.IO** for live alerts, and a **React dashboard** for real-time fleet visualization and video evidence review.

ACTIVITIES

- **Member of IEEE Student Branch TKMIT** Feb 2023 – Present
- **Member of IEDC TKMIT** Jan 2023 – Present

TECHNICAL SKILLS INTERESTS

Languages: C#, Python, Java, C, SQL, HTML/CSS

Frameworks & Tools: .NET Framework, MS SQL Server, ADO.NET, Cypress

Spoken Languages: English, Malayalam, Hindi, Arabic

Soft Skills: Problem Solving, Teamwork, Adaptability